VICTORY 2027

FORCE, WORK, POWER & ENERGY

LECTURE-4 PHYSICS

Shailendra sir



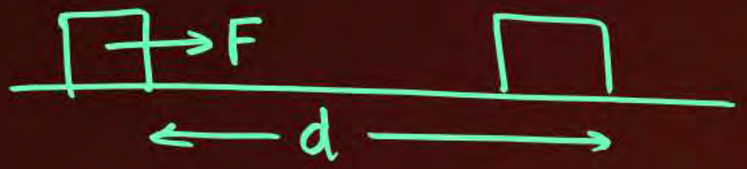
Topics To Be Covered

1. WORK
2. POWER
3. DIFFERENCE BETWEEN WORK AND POWER
4. QUESTIONS

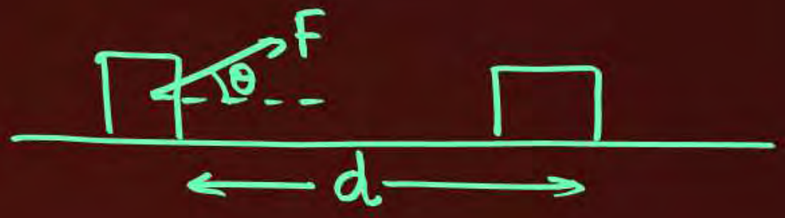

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Work

The amount of work done by a force is equal to the product of the force and the displacement in the direction of force.



$W = Fd$



$W = Fd\cos\theta$

$\cos 0^\circ = 1$

$\cos 30^\circ = \frac{\sqrt{3}}{2}$

$\cos 45^\circ = \frac{1}{\sqrt{2}}$

$\cos 60^\circ = \frac{1}{2}$

$\cos 90^\circ = 0$

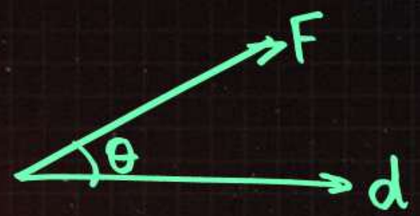
$\cos 180^\circ = -1$

$\cos\theta > 0$

$W > 0$

$\theta \rightarrow$ acute angle

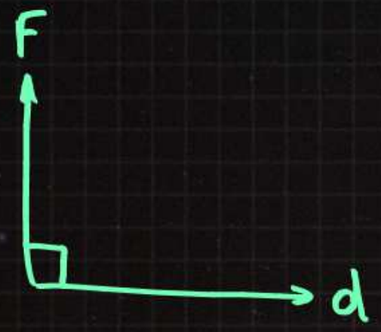
$(\theta < 90^\circ)$



$\cos\theta = 0$

$W = 0$

$\theta = 90^\circ$

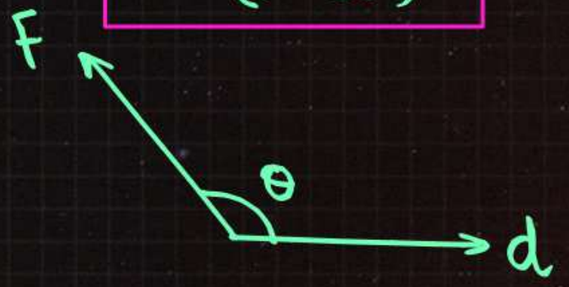


$\cos\theta < 0$

$W < 0$

$\theta \rightarrow$ obtuse angle

$(\theta > 90^\circ)$



QUESTIONS

A crane pulls up a car of mass $500kg$ to a vertical height of $4m$. Calculate the work done by the crane. $(g=10m/s^2)$

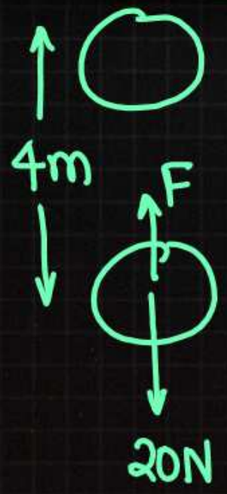
$W = Fd\cos\theta$

$= Fd\cos 0°$

$= (500 \times 10) \times 4$

$= 20000\ J$





$W_F = Fd\cos 0°$

$= 20 \times 4$

$= 80\ J$



$W_{gravity} = Fd\cos 180°$

$= 20 \times 4(-1)$

$= -80\ J$



QUESTIONS

A force of 10 N displaces a body by a distance of 2 m at an angle 60° to its own direction. Find the amount of work done.

$$W = Fd\cos\theta$$

$$= 10 \times 2 \times \cos 60^\circ$$

$$= 10 \times \cancel{2} \times \frac{1}{\cancel{2}}$$

$$= 10\ \text{J}.$$

QUESTIONS

Imp

A force of 15 N is required to pull up a body of mass 2 kg through a distance 5m along an inclined plane making an angle of 30° with the horizontal .

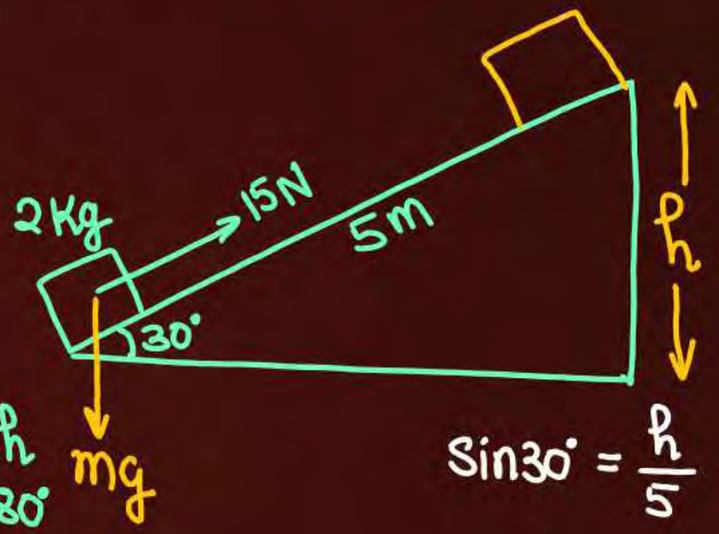
Calculate:

A. the work done by the force in pulling the body,

B. the force due to gravity on the body,

C. the work done against the force due to gravity.

Take: $g = 9.8\ ms^{-2}$



(A) $W = FdCos0°$

$= 15 \times 5$

$W = 75\ J$

(B) $F = mg$

$= 2 \times 9.8$

$F = 19.6\ N$

(C) $W = (mg)h\ Cos180°$

$W = -mgh$

$= -19.6 \times 2.5$

$W = -49\ J$

$Sin30° = \frac{h}{5}$

$h = 5\ Sin30°$

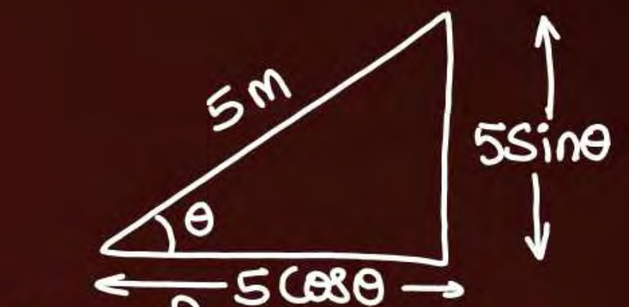
$h = 5(\frac{1}{2})$

$h = 2.5\ m$

QUESTIONS

Work has:

1. Neither magnitude nor direction
2. Only direction, no magnitude
3. Only magnitude, no direction
4. Both direction and magnitude



$\sin\theta = \frac{P}{5}$

$P = 5\sin\theta$

$\cos\theta = \frac{B}{5}$

$B = 5\cos\theta$

POSITIVE, NEGATIVE & ZERO WORK

$\theta < 90^\circ$

$W > 0$

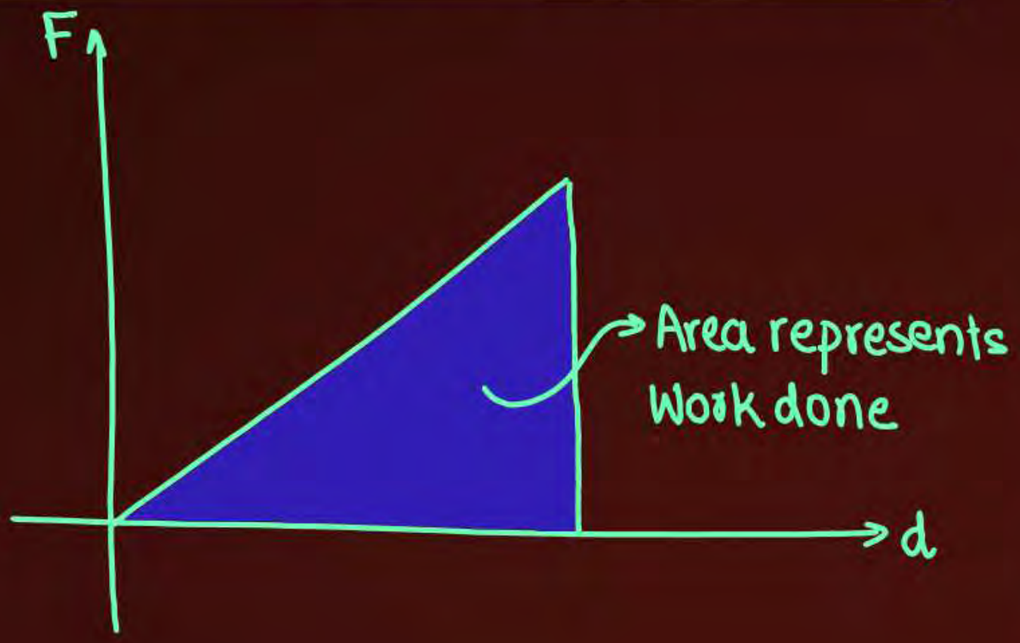
$\theta = 90^\circ$

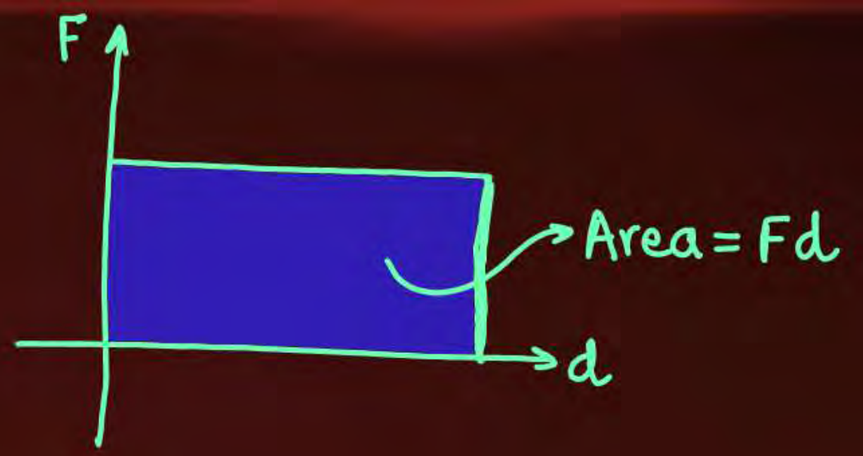
$W = 0$

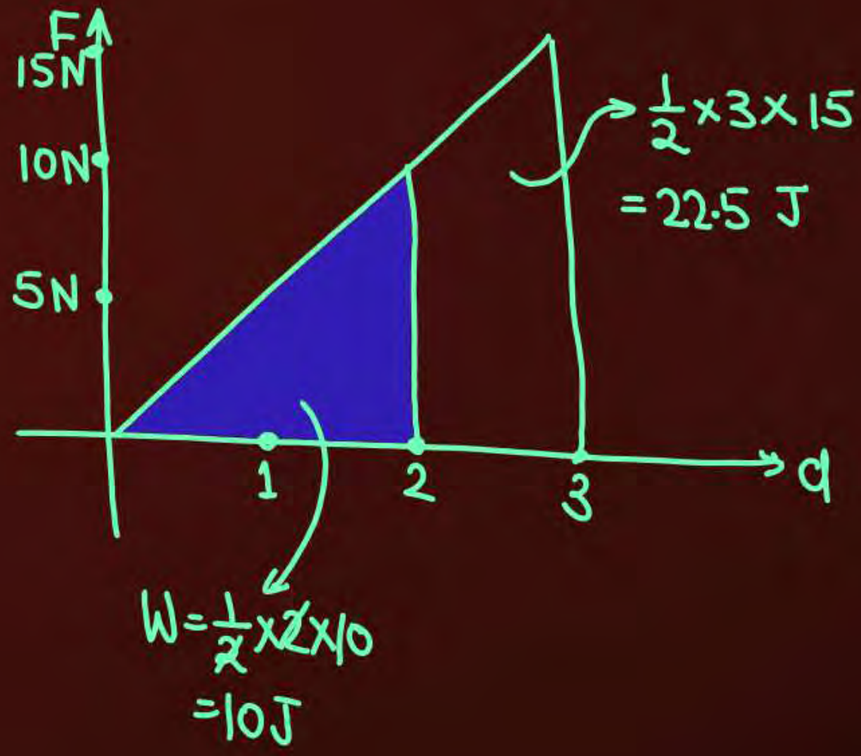
$\theta > 90^\circ$

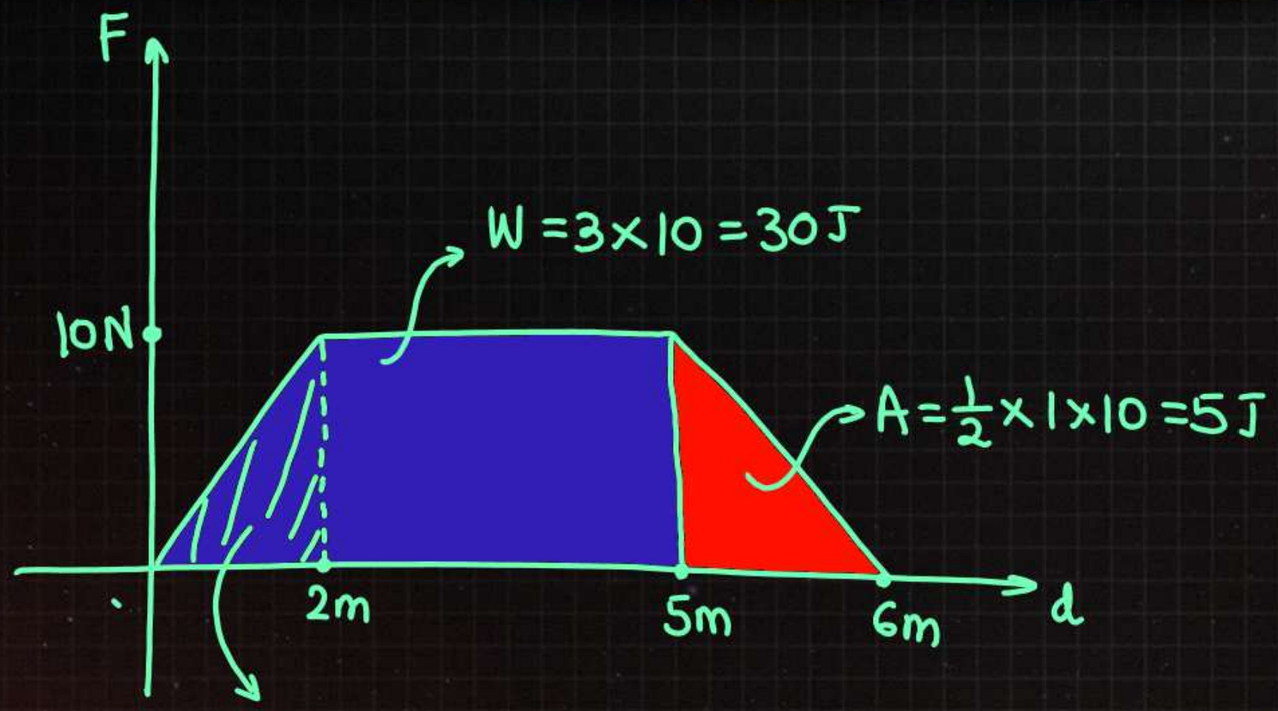
$W < 0$

WORK DONE BY A VARIABLE FORCE








F
10N
W = 3 × 10 = 30 J
A = ½ × 1 × 10 = 5 J
2m
5m
6m
d
W = ½ × 2 × 10
= 10 J

$W = F\,d$

SI: N m or Joule

CGS: dyne cm or erg

$1\ J = 1\ N\ m$

$= 10^5$ dyne 10^2 cm

$1\ J = 10^7$ erg

Gravitational Unit of Work

$W = F\,d$

↓ ↓

Kgf m

$1\ Kgf\ m = 9.8\ N\ m = 9.8\ J$

$1\ Kgf = 9.8\ N$

UNITS OF WORK

S.I. unit of work: Joule

1 joule of work is said to be done when a force of 1 newton moves a body by 1 metre in the direction of force.

Work = Force displacement

1 joule = 1 newton 1 metre

Gravitational unit of work:

1kgf m = 9.8 joule

Work = Force displacement

1kgf m = 9.8 joule

C.G.S. unit of work: erg

1 erg of work is said to be done when a force of 1 dyne moves a body by 1cm in the direction of force.

Work = Force displacement

1 erg = 1 dyne 1cm

Gravitational unit of work: 1gf cm = 980erg

Relationship between MKS and CGS unit

1 joule = 1N m; 1 joule = 10^7 erg

POWER

The power spent by a source is measured as the amount of work done per second by the source (or it is equal to the rate of doing work by the source).

$$P = \frac{W}{t} = F\frac{d}{t} = Fv$$

$$P = Fv\cos\theta$$

Scalar Quantity

SI : J/s or Watt

CGS : erg/s

UNITS OF POWER

S.I. unit: The S.I. unit of power is watt. It is abbreviated as W.

If 1 joule of work is done in 1 second, the power spent is said to be 1 watt.

$$1 \text{ watt} = \frac{1 \text{ joule}}{1 \text{ second}} = 1 \text{Js}^{-1}$$

Bigger units of power are kilowatt (kW), megawatt (MW) and gigawatt (GW), where

$$1\text{kW} = 10^3\text{W};$$
$$1\text{MW} = 10^6\text{W}$$
$$1\text{GW} = 10^9\text{W}$$

Smaller units of power are milliwatt (mW) and microwatt (μW), where

$$1\text{mW} = 10^{-3}\text{W}$$
$$1\mu\text{W} = 10^{-6}\text{W}$$

C.G.S. unit: The C.G.S. unit of power is erg per second (erg s^{-1}).

UNITS OF POWER

Horse power: It is another unit of power, largely used in mechanical engineering. It is related to the S.I. unit watt as below :

$$\boxed{1 \text{ H.P. } = 746\text{W}} = 0.746\text{kW}$$

DIFFERENCE BETWEEN WORK AND POWER

HW

	Work		Power
1.	Work done by a force is equal to the product of force and the displacement in the direction of force.	1.	Power of a source is the rate of doing work by it.
2.	Work done does not depend on time.	2.	Power spent depends on the time in which work is done.
3.	S.I. unit of work is joule (J).	3.	S.I. unit of power is watt (W).

QUESTIONS

Imp

A boy of mass 40kg climbs up a flight of 30 steps each of 20cm high in 2min and a girl of mass 30kg does the same in 1.5min.

Compare:

(i) the work done, and

(ii) the power developed by them ($g = 10\text{ms}^{-2}$).

(i) Displacement $= 30 \times \frac{20}{100} = 6\text{m}$

$W_{Boy} = mgh$

$= 40 \times 10 \times 6$

$= 2400\text{J}$

$W_{girl} = mgh$

$= 30 \times 10 \times 6$

$= 1800\text{J}$

$\frac{W_{Boy}}{W_{girl}} = \frac{2400}{1800} = \frac{4}{3}$

(ii) $P_{Boy} = \frac{W}{t} = \frac{2400}{2 \times 60} = 20\text{W}.$

$P_{girl} = \frac{W}{t} = \frac{1800}{1.5 \times 60} = 20\text{W}$

$\frac{P_{Boy}}{P_{girl}} = \frac{20}{20} = 1:1$

QUESTIONS

Calculate the power of an engine required to lift 10^5kg of coal per hour from a mine 360m deep. (Take $g = 10\text{ms}^{-2}$).

$$P = \frac{W}{t} = \frac{mgh}{t} = \frac{10^5 \times 10 \times 360}{60 \times 60} = 10^5\ \text{W}$$

QUESTIONS

Imp An electric motor of power 100W is used to drive the stirrer in a water bath. If 50% of the energy supplied to the motor is spent in stirring the water, calculate the work done on water in one minute.

$P = 100\ W$

Power supplied to motor $= 100 \times \frac{50}{100} = 50\ W$

$W = P \times t$

$= 50 \times 60$

$W = 3000\ J$

QUESTIONS

A man exerts a force of 200N in pulling a cart at a constant speed of 16ms^{-1}. Calculate the power spent by man.

$P = FV$

$= 200 \times 16$

$= 3200\ \text{W}$

$P = FV\cos\theta$

QUESTIONS

One horse power is equal to:

1. 1000 W
2. 500 W
3. 764 W
4. 746 W

Thank You